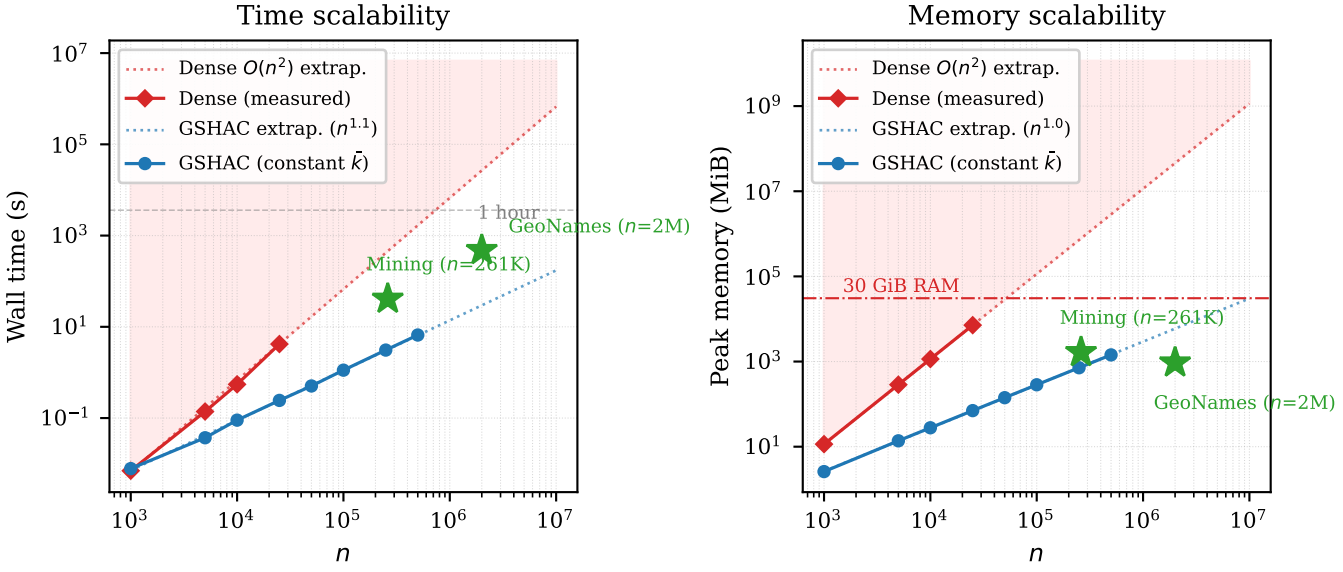
Time scalability
Wall time (s)
$10^{7}$
$10^{5}$
$10^{3}$
$10^{1}$
$10^{-1}$
Dense $O(n^2)$ extrap.
Dense (measured)
GSHAC extrap. ($n^{1.1}$)
GSHAC (constant $\bar{k}$)
1 hour
GeoNames ($n$=2M)
Mining ($n$=261K)
$10^{3}$
$10^{4}$
$10^{5}$
$10^{6}$
$10^{7}$
$n$
Memory scalability
Peak memory (MiB)
$10^{9}$
$10^{7}$
$10^{5}$
$10^{3}$
$10^{1}$
Dense $O(n^2)$ extrap.
Dense (measured)
GSHAC extrap. ($n^{1.0}$)
GSHAC (constant $\bar{k}$)
30 GiB RAM
Mining ($n$=261K)
GeoNames ($n$=2M)
$10^{3}$
$10^{4}$
$10^{5}$
$10^{6}$
$10^{7}$
$n$